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Embodied Vulnerability in Design Research

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Embodied Vulnerability in Design Research: Methods, Ethics, and Care

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Abstract

HCI and design research frequently engage with vulnerability in contexts of care, health, and life transitions, yet often overlooks how vulnerability is sensed and enacted through the body. This one-day workshop foregrounds vulnerability as an embodied phenomenon that unfolds in interaction, prior to and beyond language. Participants explore vulnerability through shared experiential inquiry: through carefully facilitated embodied activities, collective reflection, and methodological experimentation, participants will attend to embodied, sensory, and relational experiences of vulnerability in their own research practices. The workshop offers a safe, ethically attentive space in which participants experiment with translating non-verbal experiences into communicable forms such as sketches, body-maps or material traces that can inform design inquiry. We aim to cultivate a shared vocabulary, methodological sensibilities and practical techniques for engaging embodied vulnerability in design research, and lay the foundations for ongoing community building around care-oriented approaches in HCI and design.

CCS Concepts

• **Human-centered computing** → **Interaction design**; **Participatory design**.

Keywords

Vulnerability, Embodied Interaction, Participatory Design

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1 Motivation and Background

Human Computer Interaction (HCI) and Design routinely engage with vulnerability in health and wellbeing settings, such as care practices and wellbeing during life-transition [25, 34]. In these contexts, vulnerability is situated exposure to uncertainty or harm that emerges through life circumstances, relationships, environments, hindrance in abilities that disrupt life routines and moments in time rather than being a fixed attribute of a person [29, 33]. Existing research has predominantly approached vulnerability through verbal and cognitive lenses that rely on participants' reflective narration and linguistic articulation of experience, for example, semi-structured interviews, surveys, or think-aloud protocols. These approaches have value, but overshadow how vulnerability is felt and negotiated in and through the body. An embodied perspective seeks access to pre-verbal, sensory-affective, and relational experiences, that is, not only what people say, but what they do and feel together [14, 20]. In HCI, this shift aligns with participatory design research [13, 16, 23, 35], embodied interaction [12, 24, 30, 31] and soma design [22, 27, 28], which argue for moving beyond symbolic, language-oriented methods to experiential, first-person and relational inquiry [8, 15].

A focus on embodiment makes visible what verbal accounts can miss, opening new possibilities for design research and practice [8]. For instance, during life transitions when young adults are moving abroad, vulnerability may manifest as disrupted sleep rhythms, a hunched posture in crowded spaces, or moments of embodied relief when encountering a familiar smell [3, 10, 11, 17]. In these instances, researchers also become vulnerable: navigating uncertainty, ethical tensions, and their own responses as roles and environments shift [1, 4, 32]. For example, a person with dementia may feel safe and grounded in a familiar home setting, while the researcher becomes vulnerable to navigate uncertainty, responsibility, and ethical tensions [26]. Across these contexts, lived experiences of vulnerability are often pre-verbal, and hence attending to such experiential notions is essential for designing with care, accessibility, and relational attunement. These examples illustrate that vulnerability arises and unfolds in interaction, fluctuating between people, moments, and settings.

This workshop contributes to ongoing conversations in HCI and design about vulnerability [5, 6, 18], ethics [2, 9, 34], and care [7, 21, 36], by foregrounding embodied reflection [19] and shared experiential inquiry, while offering alternative ways of knowing vulnerability beyond primarily verbal or representational accounts. We invite participants to collectively explore how vulnerability is sensed, negotiated, and transformed by intentionally engaging with vulnerability within a safe and attentive space. By doing so, we aim to allow an understanding of vulnerability to emerge, whether through participants reflecting on their own vulnerability as researchers, the vulnerability of those they design with and for, or the interactional dynamics through which vulnerability is co-produced in design process.

This workshop seeks to explore how vulnerability might be understood as something that is inhabited and encountered through embodied and relational experience. Rather than treating vulnerability as a fixed condition, we invite participants to examine how it unfolds within oneself, between people, and through interactions over time. Vulnerability is not necessarily always something to be avoided or minimized. In many design practices, particularly those engaging with fundamental human well-being, transitional experiences, or contexts of care, vulnerability may be intentionally entered, invited, or even produced within ethically held spaces. During design practices, researchers may expose themselves, or invite participants to moments of uncertainty, discomfort, or loss of control in order to access forms of insight, empathy, or relational closeness that would otherwise remain unavailable. From this perspective, vulnerability is not merely fragile or risky, but potentially generative and productive, characterized by the ongoing ebbs and flows rather than a fixed state as vulnerability is experienced.

Yet, attending to vulnerability in this way poses a methodological and epistemic challenge. This workshop addresses this challenge by foregrounding attention to what participants notice in their bodies, in their interactions and relations with others, and in the context of interaction, especially when such experiences of vulnerability are challenging to articulate. The workshop creates a space for participants to reflect on and share experiences of vulnerability from their own research practice, with particular emphasis on what was strongly felt but not easily expressed. It begins with a facilitated experiential embodied activity that invites participants into a mild, situated experience of vulnerability within a safe and supported setting. From there, the workshop unfolds responsively, attending to, shaping, and being shaped by participants' experiences of vulnerability.

2 Workshop Themes

(1) **Exploring Vulnerability as Lived, Embodied, and Relational**: We will approach vulnerability as something sensed in and through bodies and relations, often before it can be named. Participants will attend closely to subtle bodily cues, felt sensations, and relational atmospheres as sites of knowledge. We will examine how vulnerability appears differently when inhabiting roles such as researcher, participant, designer, carer, or peer and how it is co-produced through the setting, the tools in use, and the tacit norms that shape interaction. We will explore vulnerability not only as risk but as a resource that can bring researchers and participants closer to what understanding lived experiences. Participants may share, through embodied means such as enactment, moments where leaning into uncertainty, discomfort, or loss of control made possible a new kind of insight, empathy, or relational attunement, for example, in transitional experiences, matters of care, or work touching human needs and well-being. We will carefully trace how mild, intentionally held vulnerability can open access to stories, sensations, and perspectives that remain unavailable under more controlled conditions. The aim is to understand how vulnerability's unfolding quality can be productive for design inquiry without necessarily assuming a need for resolution.

(2) **Promoting Safety, Consent and Care**: Attending to what is strongly experienced and felt requires spaces that are safe, flexible, and responsive. This workshop will focus on practices that make such attending possible: informed and ongoing consent, clear opt-in and opt-out pathways, grounding and debriefing, and aftercare that acknowledges lingering affects. For instance, accessibility and trauma-informed considerations will be foregrounded, recognizing that embodied work may exceed what participants can easily anticipate or articulate. Ethics here is a relational practice, sustained moment-by-moment rather than guaranteed by forms alone. This workshop further aims to explore how vulnerability is rarely linear. It oscillates between for example closeness and distance, clarity and confusion. We will make space for those movements and their implications for design judgment. We consider what is revealed about boundaries, difference, and care.

(3) **Methodological Responsiveness**: Methodological responsiveness means allowing inquiry to be shaped by what is felt in the moment, rather than by a pre-fixed goal. In this workshop, participants will explore how methods can adapt to emergent embodied sensations and interactional dynamics of vulnerability. The aim is to scaffold attentiveness to embodied knowing and to build capacity for "staying with" the pre-verbal experiences of vulnerability. Responsiveness also includes knowing when not to capture, when silence, slowness, or pause is the most ethical and informative method. While felt experiences often resist language, design research relies on forms of articulation that can be shared, reflected upon, and built into practice. We will explore ways to carefully translate the tacit into communicable forms, such as body maps, sketches or material traces. Participants will experiment with representational strategies, developing shared vocabularies that are descriptive without being reductive. The goal is to cultivate forms of articulation that support collective sense-making and ethical accountability, while keeping the felt experience of vulnerability present within design conversations and decisions.

2.1 Workshop Goals

The goal of the workshop is to nurture a safe space toward:

- Creating a supportive interdisciplinary forum for researchers, designers, and practitioners to examine vulnerability as it is lived, sensed, and negotiated in design practice.
- Surfacing and discussing experiences of vulnerability from participants' own work, including moments of discomfort, uncertainty, attunement, and estrangement.
- Offering hands-on, embodied engagement with vulnerability through carefully facilitated experiential activities that foreground safety, consent, and care.
- Exploring vulnerability as a dynamic and generative process, rather than a fixed state to be avoided, particularly in contexts involving care, transition, and closeness to fundamental human needs. Reflecting on ethical and methodological considerations involved in "producing" or inviting vulnerability within research and design encounters.
- Articulating implications for design research practice, including how embodied and relational insights can inform methods, reflexivity, and ethical decision-making.
- Contributing to longer-term methodological and ethical development in HCI and design research by shaping shared language, sensibilities, and practices for attending to embodied vulnerability beyond the workshop, informing future studies, collaborations, and pedagogical approaches.

2.2 Anticipated Outcomes

Through the embodied, experiential activities, collective reflection, and by drawing on the situated expertise of facilitators and participants, the anticipated outcomes of the workshop include:

- A shared understanding of vulnerability as an embodied and relational phenomenon, articulated through participants' lived experiences rather than abstract definitions.
- A comparative view of how vulnerability is engaged across design contexts, such as care, health, life transitions, participatory design, and sensitive fieldwork.
- Critical reflections on the potentials and tensions of intentionally engaging vulnerability in design research, including ethical, methodological, and relational implications.
- Emergent practices and vocabularies for working with felt, pre-verbal, and difficult-to-articulate experiences in design inquiry.
- Foundations for continued collaboration, such as a joint publication, workshop report, or methods-oriented contribution to HCI/design venues.
- An emerging community of researchers and practitioners interested in embodied, affective, and care-oriented approaches to vulnerability in design.

References

[1] Madeline Balaam, Rob Comber, Rachel E Clarke, Charles Windlin, Anna Ståhl, Kristina Höök, and Geraldine Fitzpatrick. 2019. Emotion work in experience-centered design. In *Proceedings of the 2019 CHI conference on human factors in computing systems*. 1–12.

[2] Barry Brown, Alexandra Weilenmann, Donald McMillan, and Airi Lampinen. 2016. Five provocations for ethical HCI research. In *Proceedings of the 2016 CHI conference on human factors in computing systems*. 852–863.

[3] Keith B Burt and Ann S Masten. 2010. Development in the transition to adulthood: Vulnerabilities and opportunities. *Young adult mental health* (2010), 5–18.

[4] Caterina Calefato, Luca Cattani, Judith L Charlton, Sharon Cook, Eva Eriksson, Chiara Ferrarini, Elaine Gosling, Colette Nicolle, and Olof Torgersson. 2014. Training designers for vulnerable generations: a quest for a more inclusive design. *Interaction Design and Architecture (s)* 21, 1 (2014), 25–36.

[5] Stevie Chancellor, Nazanin Andalibi, Lindsay Blackwell, David Nemer, and Wendy Moncur. 2019. Sensitive research, practice and design in HCI. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–8.

[6] Carla Cipolla. 2018. Designing for vulnerability: Interpersonal relations and design. *She Ji: The Journal of Design, Economics, and Innovation* 4, 1 (2018), 111–122.

[7] Anna Croon. 2022. Thinking with care in human–computer interaction. *Feminist Theory* 23, 2 (2022), 232–246.

[8] Humphrey Curtis and Timothy Neate. 2024. Beyond Repairing with Electronic Speech: Towards Embodied Communication and Assistive Technology. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–12.

[9] Rachael Garrett, Kristina Popova, Claudia Núñez-Pacheco, Thorhildur Asgeirsdottir, Airi Lampinen, and Kristina Höök. 2023. Felt ethics: Cultivating ethical sensibility in design practice. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–15.

[10] Oliver L Haimson, Bryan Semaan, Brianna Dym, Joey Chiao-Yin Hsiao, Daniel Herron, and Wendy Moncur. 2019. Life transitions and social technologies: Research and design for times of life change. In *Companion Publication of the 2019 Conference on Computer Supported Cooperative Work and Social Computing*. 480–486.

[11] Daniel Herron, Nazanin Andalibi, Oliver Haimson, Wendy Moncur, and Elise Van Den Hoven. 2016. HCI and sensitive life experiences. In *Proceedings of the 9th Nordic Conference on Human-Computer Interaction*. 1–3.

[12] Mads Hobye and Jonas Löwgren. 2011. Touching a stranger: Designing for engaging experience in embodied interaction. *International Journal of Design* 5, 3 (2011), 31–48.

[13] Elise Hodson, Annukka Svanda, and Nastaran Dadashi. 2023. Whom do we include and when? Participatory design with vulnerable groups. *CoDesign* 19, 4 (2023), 269–286.

[14] Sarah Homewood, Marika Hedemyr, Maja Fagerberg Ranten, and Susan Kozel. 2021. Tracing conceptions of the body in HCI: From user to more-than-human. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–12.

[15] Kristina Höök, Sara Eriksson, Marie Louise Juul Søndergaard, Marianela Ciolfi Felice, Nadia Campo Woytuk, Ozgun Kilic Afsar, Vasiliki Tsaknaki, and Anna Ståhl. 2019. Soma design and politics of the body. In *Proceedings of the Halfway to the Future Symposium 2019*. 1–8.

[16] Maarten Houben, Nena Van As, Nitin Sawhney, David Unbehaun, and Minha Lee. 2023. Participatory design for Whom? Designing conversational user interfaces for sensitive settings and vulnerable populations. In *Proceedings of the 5th International Conference on Conversational User Interfaces*. 1–4.

[17] Armağan Karahanoğlu, Laia Turmo Vidal, Daniel Harrison, Jamie Steane, Tina Ekhtiar, Teresa Almeida, Anna Vallgårda, and Madeline Balaam. 2024. Life in transitions: the role of technology in supporting well-being in the heart of change. In *Adjunct Proceedings of the 2024 Nordic Conference on Human-Computer Interaction*. 1–5.

[18] Calvin A Liang, Sean A Munson, and Julie A Kientz. 2021. Embracing four tensions in human-computer interaction research with marginalized people. *ACM Transactions on Computer-Human Interaction (TOCHI)* 28, 2 (2021), 1–47.

[19] Lian Loke and Toni Robertson. 2013. Moving and making strange: An embodied approach to movement-based interaction design. *ACM Transactions on Computer-Human Interaction (TOCHI)* 20, 1 (2013), 1–25.

[20] Kaarina Mononen. 2019. Embodied care: affective touch as a facilitating resource for interaction between caregivers and residents in a care home for older adults. *Linguistics Vanguard* 5, s2 (2019).

[21] Adrian Petterson, Jocelyn Mattka, and Priyank Chandra. 2025. Expanding Care Conceptualizations: An Integrative Literature Review of Care in HCI. In *Proceedings of the ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies*. 481–503.

[22] Kristina Popova, Rachael Garrett, Claudia Núñez-Pacheco, Airi Lampinen, and Kristina Höök. 2022. Vulnerability as an ethical stance in soma design processes. In *Proceedings of the 2022 CHI conference on human factors in computing systems*. 1–13.

[23] Sneha Raman and Tara French. 2022. Participatory design in sensitive contexts: A proposal for a conceptual framework. *The Design Journal* 25, 5 (2022), 752–767.

[24] Maja Fagerberg Ranten, Mads Hobye, Troels Andreasen, Karen Eide Bøen, and Lise Aagaard Knudsen. 2025. Designing for Einfühlung: Strategies for Embodied Sensitivity in Interactive Spaces. In *IFIP Conference on Human-Computer Interaction*. Springer, 289–299.

[25] Pedro Sanches, Axel Janson, Pavel Karpashevich, Camille Nadal, Chengcheng Qu, Claudia Daudén Roquet, Muhammad Umair, Charles Windlin, Gavin Doherty, Kristina Höök, et al. 2019. HCI and Affective Health: Taking stock of a decade of studies and charting future research directions. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–17.

[26] Emma Sherry. 2013. The vulnerable researcher: Facing the challenges of sensitive research. *Qualitative Research Journal* 13, 3 (2013), 278–288.

[27] Anna Ståhl, Madeline Balaam, Rob Comber, Pedro Sanches, and Kristina Höök. 2022. Making New Worlds–Transformative Becomings with Soma Design. In *Proceedings of the 2022 Chi conference on human factors in computing systems*. 1–17.

[28] Anna Ståhl, Vasiliki Tsaknaki, and Madeline Balaam. 2021. Validity and rigour in soma design-sketching with the soma. *ACM Transactions on Computer-Human Interaction (TOCHI)* 28, 6 (2021), 1–36.

[29] Xinru Tang, Gabriel Lima, Jiang Jiang, Li, Lucy Simko, and Yixin Zou. 2025. Beyond "Vulnerable Populations": A Unified Understanding of Vulnerability From A Socio-Ecological Perspective. *Proc. ACM Hum.-Comput. Interact.* 9, 2, Article CSCW037 (May 2025), 30 pages. doi:10.1145/3710935

[30] Daniel Tetteroo, Iris Soute, and Panos Markopoulos. 2013. Five key challenges in end-user development for tangible and embodied interaction. In *Proceedings of the 15th ACM on International conference on multimodal interaction*. 247–254.

[31] Oscar Tomico and Danielle Wilde. 2015. Soft, embodied, situated & connected. In *Proceedings of the 17th International Conference on Human-Computer Interaction with Mobile Devices and Services Adjunct*. 1179–1186.

[32] Anupriya Tuli, Azra Ismail, Karthik S Bhat, Pushpendra Singh, and Neha Kumar. 2023. "Information-Backward but Sex-Forward": Navigating Masculinity towards Intimate Wellbeing and Heterosexual Relationships. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) *(CHI '23)*. Association for Computing Machinery, New York, NY, USA, Article 39, 16 pages.

[33] Janna van Grunsven, Naomi Jacobs, Bart A. Kamphorst, Michaela Honauer, Bouke van Balen, Caroline Bollen, Matthew Dennis, Anna Puzio, and M. Birna van Riemsdijk. 2025. Three Perspectives on Human Vulnerability: Enriching Responsible Computing Design Processes1. *ACM J. Responsib. Comput.* 2, 4, Article 15 (Oct. 2025), 21 pages. doi:10.1145/3748731

[34] Jenny Waycott, Cosmin Munteanu, Hilary Davis, Anja Thieme, Stacy Branham, Wendy Moncur, Roisin McNaney, and John Vines. 2017. Ethical encounters in HCI: Implications for research in sensitive settings. In *Proceedings of the 2017 CHI conference extended abstracts on human factors in computing systems*. 518–525.

[35] Jenny Waycott, Greg Wadley, Stefan Schutt, Arthur Stabolidis, and Reeva Lederman. 2015. The Challenge of Technology Research in Sensitive Settings: Case Studies in'ensitive HCI'. In *Proceedings of the Annual Meeting of the Australian Special Interest Group for Computer Human Interaction*. 240–249.

[36] Cara Wilson, Larissa Pschetz, Billy Dixon, Sue J Lewis, Joe Revans, and John Vines. 2024. Creating Resources for Designing with and for Care Ecologies in HCI. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference*. 3161–3178.